Parktown Boys' High School

Geography Department

Grade 12 June Exam

Examiner: Mr A Meintjes — Time: 180 Minutes

Moderator: E. Prideaux-Brune (KES) & Mr D Grace — Total: 150 Marks

Date: 13 June 2024

INSTRUCTIONS AND INFORMATION

1. This question paper consists of TWO SECTIONS.

   SECTION A

   QUESTION 1: CLIMATE AND WEATHER (40 MARKS)

   QUESTION 2: GEOMORPHOLOGY (40 MARKS)

   QUESTION 3 : SETTLEMENT GEOGRAPHY (40 MARKS)

   SECTION B

   QUESTION 4: GEOGRAPHICAL SKILLS AND TECHNIQUES (30 MARKS)

   Answer ALL FOUR questions.

2. The question paper consists of FOUR questions.
3. ALL diagrams are included in the question paper.

4. Where possible, illustrate your answers with labelled diagrams.

5. Leave a line between subsections answered.

6. Start EACH section at the top of a NEW page.

7. Number your answers correctly according to the numbering system used in this question paper

8. Do NOT write in the margins of your ANSWER BOOK.

9. Write neatly and legibly.

10. In SECTION B you are provided with a 1 : 50 000 topographic map (2931 CA VERULAM) and an orthophoto map (2931 CA 11 VERULAM) of a part of the mapped area.

11. Show ALL calculations and use the formulae provided, where applicable. Marks will be allocated for these.

12. Indicate the unit of measurement in the final answer of calculations,
e.g. 10 km; 2,1 cm.

13. You may use a non-programmable and a magnifying glass.

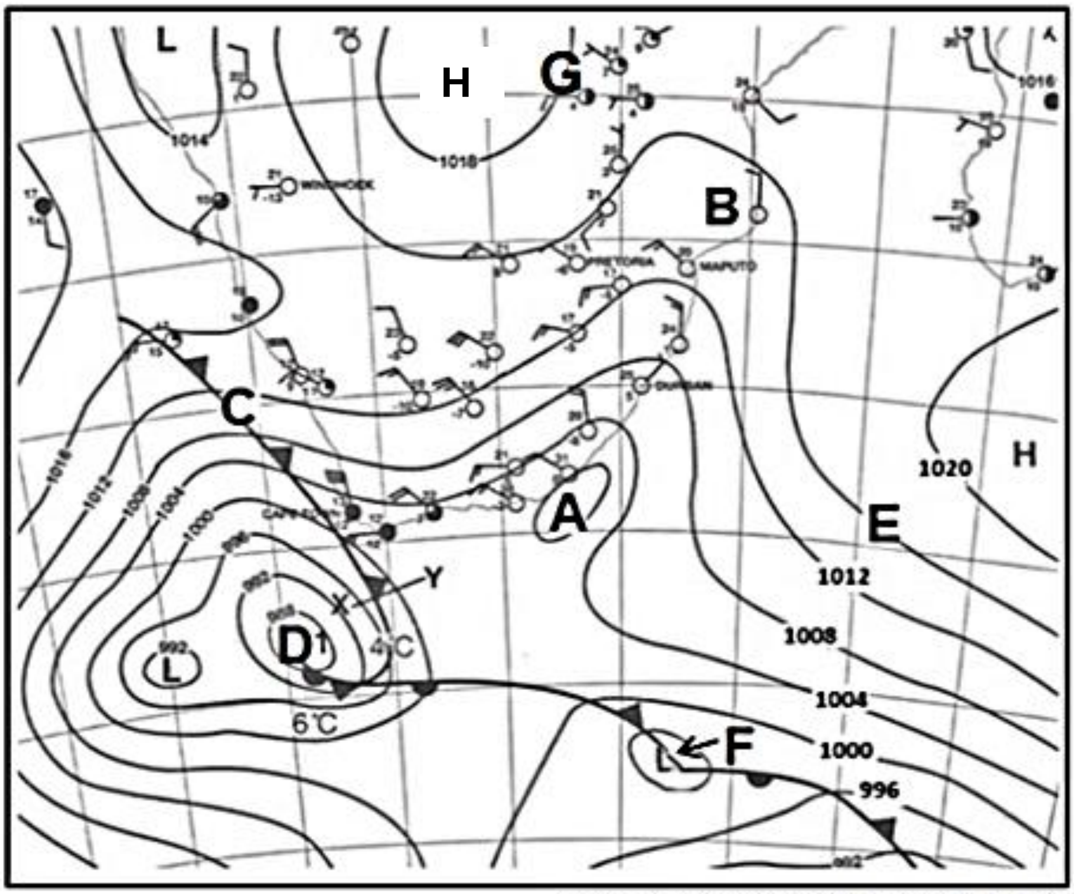
14. The area demarcated in RED and BLACK on the topographic map represents the area covered by the orthophoto map.

SECTION A: THEORY

QUESTION 1 **CLIMATOLOGY**

Question 1.1 **Synoptic charts**

Choose the correct word(s) from those given in brackets. Write only the word(s) next to the question numbers (1.1.1 to 1.1.5) in the ANSWER BOOK.



[Source: South African Weather Bureau]

1.1.1 Pressure cell **A** is a (Cut off / Coastal) low pressure.

1.1.2 Area **B** is experiencing (Northerly/ Southerly) winds.

1.1.3 The system at **F** is a (Tropical Cyclone / Mid latitude cyclone).

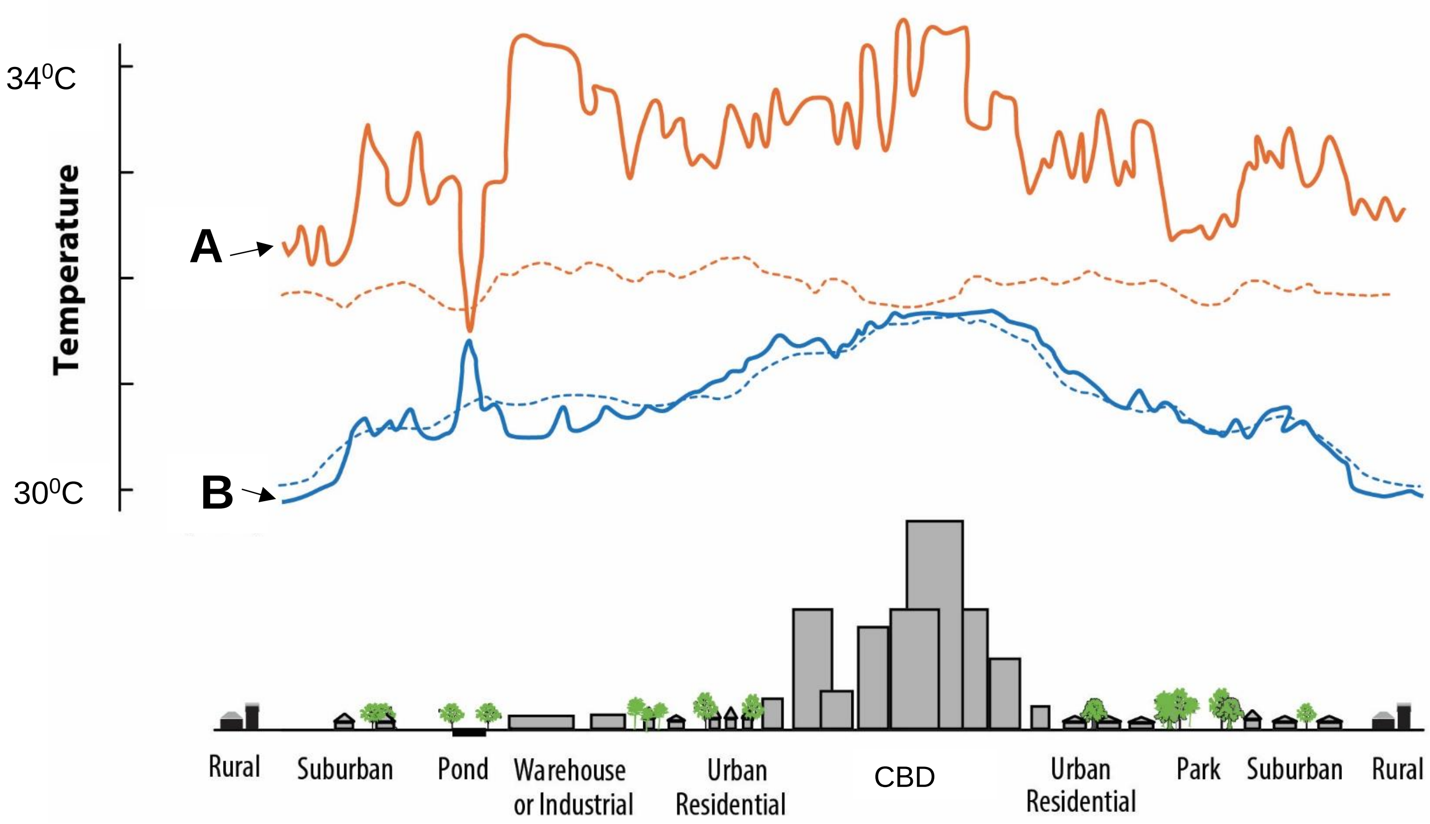
1.1.4 The pressure cell at **H** is a (Thermal Low pressure / Kalahari High pressure).

1.1.5 The synoptic chart is representing a (Summer / Winter) season.

(5x1) (5)

Question 1.2 **City climates**

Choose which graph best matches the description given. Write down the question number (1.2.8) and A or B only (1.2.8 A)



1.2.1 Graph …shows the time when artificial heat is mostly given off.

1.2.2 There is an increase in this graph due to water holding heat longer.

1.2.3 Roof top gardens and additional vegetation will cause a huge difference in which graph?

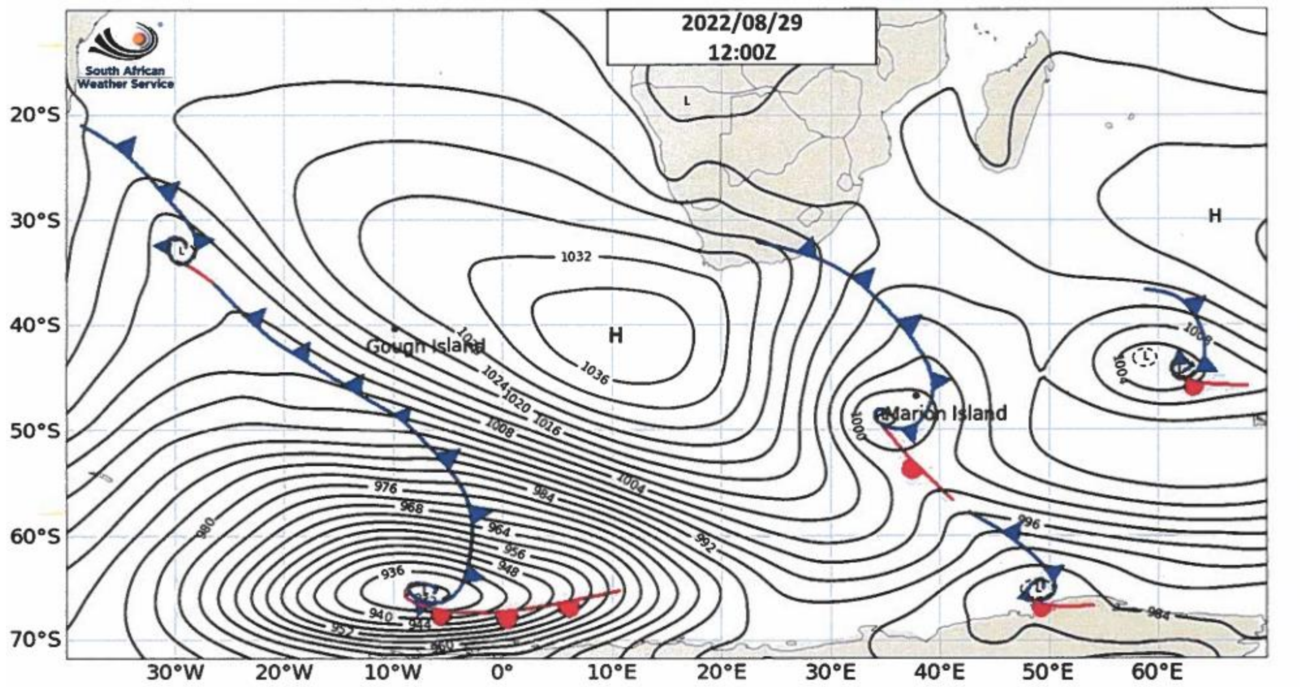
1.2.4 A Kalahari high-pressure cell has the most effect on which graph?

1.2.5 Which graph represents an urban heat island?

(5x1) (5)

[10]

Question 1.3 **Mid-Latitude Cyclones**



22 die as snowstorms hit South Africa

Chris McGreal, Johannesburg (modified for the June exam)

Parts of South Africa were declared a disaster area yesterday after the worst snow in decades left 22 people dead, destroyed thousands of homes and killed large numbers of livestock.

At least four people froze to death as snow up to 1metre deep in the Eastern Cape and KwaZulu-Natal cut off towns as well as the only ski resort in sub-Saharan Africa, where hundreds are still awaiting rescue. Others died after their vehicles were washed away by swollen rivers.

Among the dead are two herders found frozen on mountains at Lady Frere. Two other people, one in his twenties and the other 77, froze to death in the town of Elliot, among the worst hit by the snow. It was cut off for three days.

At the weekend, disaster workers rescued more than 100 people trapped in their cars by the snow, some of them for more than a day. The motorway between Johannesburg and Durban was closed by the weather.

Many farmers have no idea whether their livestock has survived. Rory O'Moore, manager of Agri Eastern Cape, was pessimistic: "We are experiencing a disaster. It was the worst snowstorm in 70 years, so we expect the worst."

The South African weather service says the worst is not over. A new cold front is expected on Thursday.

1.3.1 Name the weather phenomenon that a cold front is part of. (1x1) (1)

1.3.2 Give one phrase from the passage that states that snowstorms are not a common occurrence in South Africa. (1x1) (1)

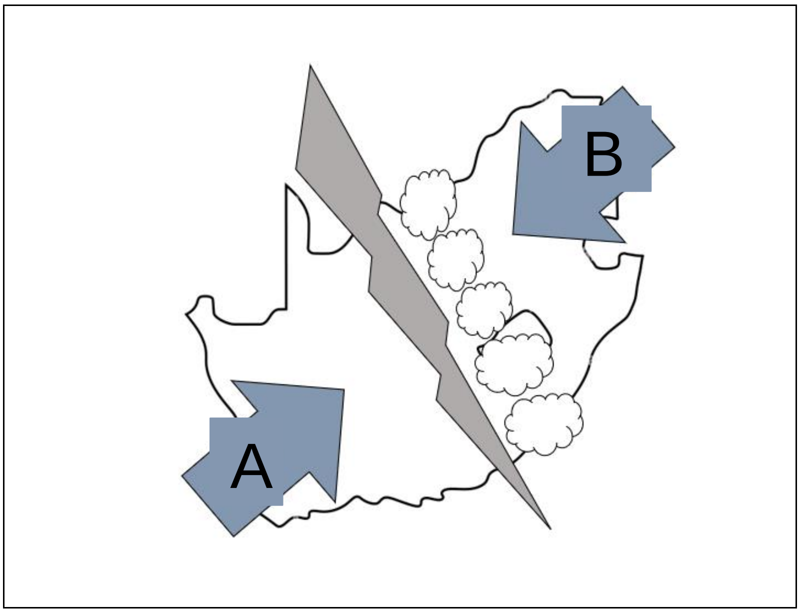
1.3.3 Where does the extreme cold air of this system originate (come from)? (1x1) (1)

1.3.4 Fully explain how a cold front causes snow to develop over South Africa? (2x2) (4)

1.3.5 In a paragraph of approximately EIGHT lines, discuss how a cold front can increase food insecurity in South Africa. (4x2) (8)

[15]

Question 1.4 The figure shows the development of a moisture front.



1.4.1 What is a *moisture front*? (1x2) (2)

1.4.2 In which season does a moisture front mainly develop? (1x1) (1)

1.4.3 Name the pressure cells that cause the winds at **A** and **B** respectively. (2x1) (2)

1.4.4 How does the air from pressure cells **A** and **B** influence the development of the moisture front? (2x2) (4)

1.4.5 Explain how this system can be both a curse and a blessing to farmers. (3x2) (6)

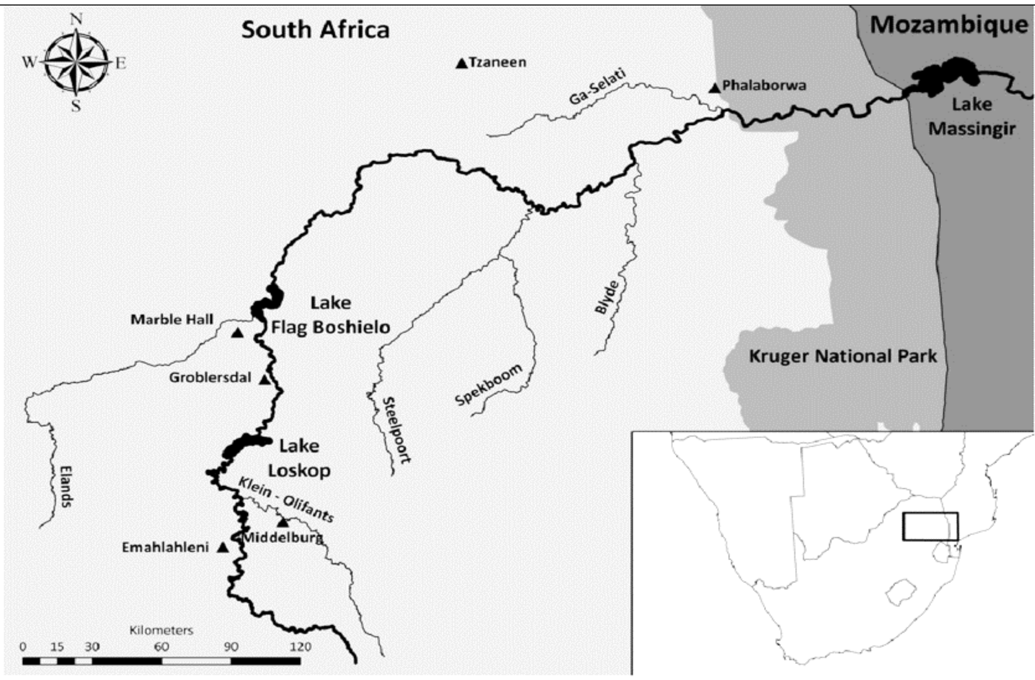
[15]

QUESTION 1 TOTAL: 40 MARKS

QUESTION 2 **GEOMORPHOLOGY**

Question 2.1. **Stream order**

Refer to the map showing the Olifants river system. Complete the statements in COLUMN A with the options in COLUMN B. Write only **X** or **Y** next to the question numbers (2.1.1 to 2.1.5) in the ANSWER BOOK, e.g. 2.1.6 Y.



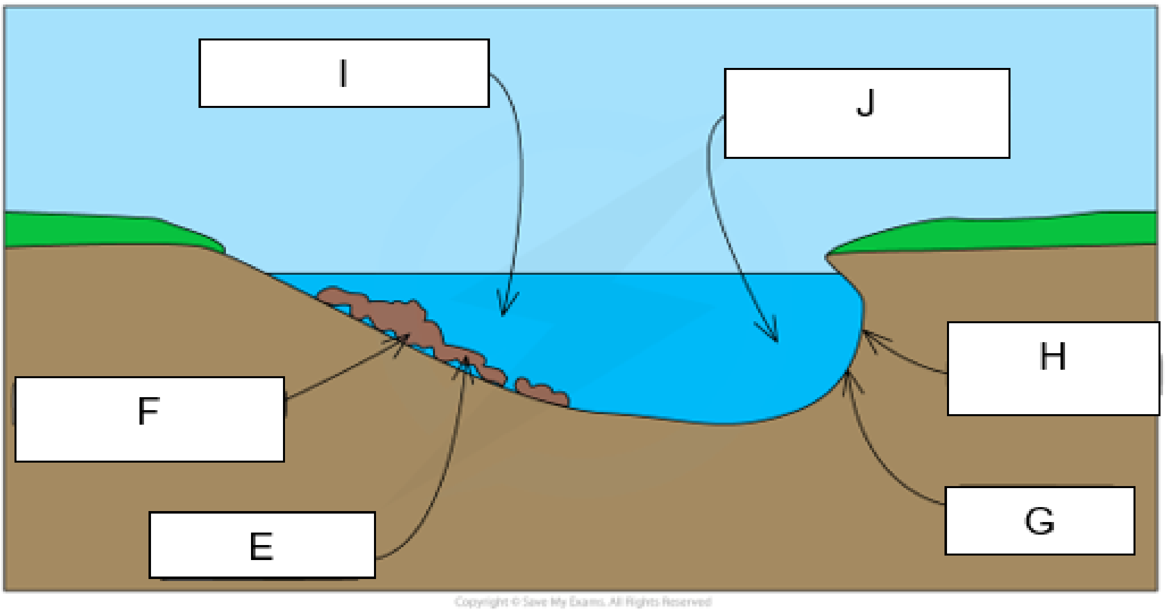
https://www.researchgate.net/figure/Map-of-the-Olifants-River-catchment-in-South-Africa

COLUMN A	COLUMN B
2.1.1 The stream order for the Olifants river as it flows into the Kruger National Park is a...	X. 2nd order stream Y. 3rd order stream
2.1.2 During times of drought the stream order will...	X. ...increase Y. ...decrease
2.1.3 The lower the stream order the …	X. ... smaller the drainage basin Y. …bigger the drainage basin
2.1.4 The higher the stream order the …	X. ... steeper the gradient of the river. Y. … more gentle the gradient of the stream
2.1.5 The Steelpoort and Spekboom tributaries are divided by …	X. ... the high lying area called a watershed. Y. ... the high lying area called an interfluve.

(5x1) (5)

Question 2.2 **Meanders**

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (**A–D**) next to the question numbers (2.2.1 to 2.2.5) in the ANSWER BOOK, e.g. 2.2.6 D. Points **E - J** show different characteristics of the cross section of a meander.



2.2.1 The course of the river where one would find meanders forming.

A Upper course / Lower course
B Middle course / Lower course
C Middle course / Upper course
D All of the above

2.2.2 Name the process that is taking place at **E**.

A Headward erosion
B Deposition
C Erosion
D Infiltration

2.2.3 Name the process that is taking place at **G**.

A Headward erosion
B Deposition
C Erosion
D Infiltration

2.2.4 The river will flow the fastest at point ___ and slowest at point ___.

A I and J

B J and I

C None of the above. Fastest point is in the middle.

D None of the above. The speed of the river is uniform all over.

2.2.5 Once the meander neck erodes through, it forms a/an___ and when the feature dries out it becomes a/an ___

i meander pool.

ii ox bow lake.

iii dry ox bow lake.

iv meander scar.

A i and iv

B ii and iii

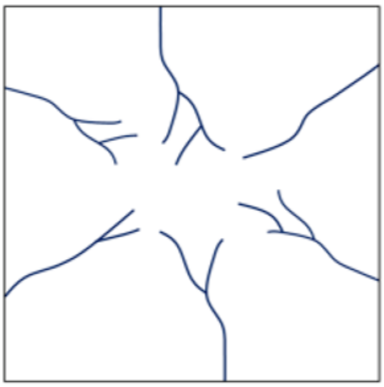
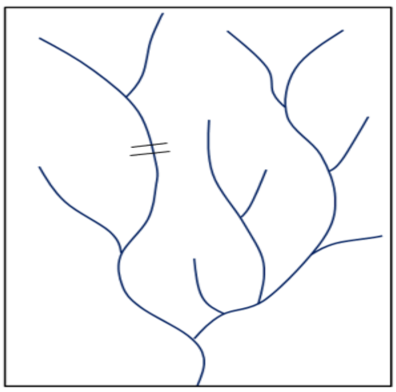
C ii and iv

D i and iii

(5x1) (5)

Question 2.3 **Drainage basins and characteristics.**

Refer to the diagrams below showing drainage patterns and answer the questions that follow.

 A	 B
 Waterfall	

2.3.1 Identify drainage patterns **A** and **B** respectively. (2x1) (2)

2.3.2 Differentiate between the underlying rock structure of drainage patterns **A** and **B** respectively. (2x2) (4)

2.3.3 Look at pattern **B**. the river shows an *ungraded* profile, and answer the following:

a) Give ONE reason to support the statement above. (1x1) (1)

b) Draw a longitudinal profile of river **B**. (1x1) (1)

c) Discuss the processes that would need to take place for the profile to become a graded profile. (2x2) (4)

2.3.4 Drainage pattern **B** shows a high drainage density.

a) Define the term *drainage density.* (1x2) (2)

b) What is the relationship between drainage density and gradient? (1x1) (1)

[15]

Question 2.4 **River Capture**

Refer to the figure, showing the Kouga River being captured by the Gamtoos river in the Eastern Cape.

2.4.1 Define the term *river capture*. (1x2) (2)

2.4.2 The process that caused the Kouga River to be captured by a tributary of the Gamtoos river, resulting in the Diep River, was abstraction.

Explain the process of abstraction. (1x2) (2)

2.4.3 Draw a sketch of the river capture shown in the figure between the Kouga River and Diep River. On your sketch show the following:

Wind gap

Misfit stream

Captor stream (3x1) (3)

2.4.4 The Captor stream becomes rejuvenated.

a) Define the term river *rejuvenation.* (1x2) (2)

b) List TWO landforms that can occur due to rejuvenation. (2x1) (2)

c) Discuss what effect river capture will have on the volume of water along the Diep river and how this difference in volume will affect the commercial farmers along the river. (2x2) (4)

[15]

QUESTION 2 TOTAL: 40 MARKS

QUESTION 3 **Settlement**

Question 3.1.

Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

3.1.1. In rural areas industrialisation and the modernisation of agricultural practices has led to:

A. Voluntary resettlement

B. Rural-urban migration

C. Basic need philosophy

D. Formal resettlement (1)

3.1.2. Which of the following factors is NOT a site factor to consider in the development of a settlement

A. Availability of water

B. Flat gradients

C. Central point of trade

D. A natural harbour (1)

3.1.3. Which is the smallest settlement?

A. Hamlet

B. Village

C. Isolated farmstead

D. City (1)

3.1.4. Which of the following would be an obstacle faced by rural-urban migrants?

A. Lack of capital

B. New lifestyles and associated costs

C. ONLY A

D. A and B (1)

3.1.5. Rural economic development is hindered because of:

A. These areas are often remote

B. These areas are seen as poor investments

C. Many workers suffer from diseases

D. All of the above (1)

Question 3.2.

State whether the following statements are **TRUE** or **FALSE**. Write only TRUE or FALSE next to the corresponding number in your answer book.

3.2.1. Rural areas are multifunctional in nature.

3.2.2. The situation of a settlement refers to its location in relation to its surrounding features.

3.2.3. Rural farming settlements is most prevalent in more economically developed countries.

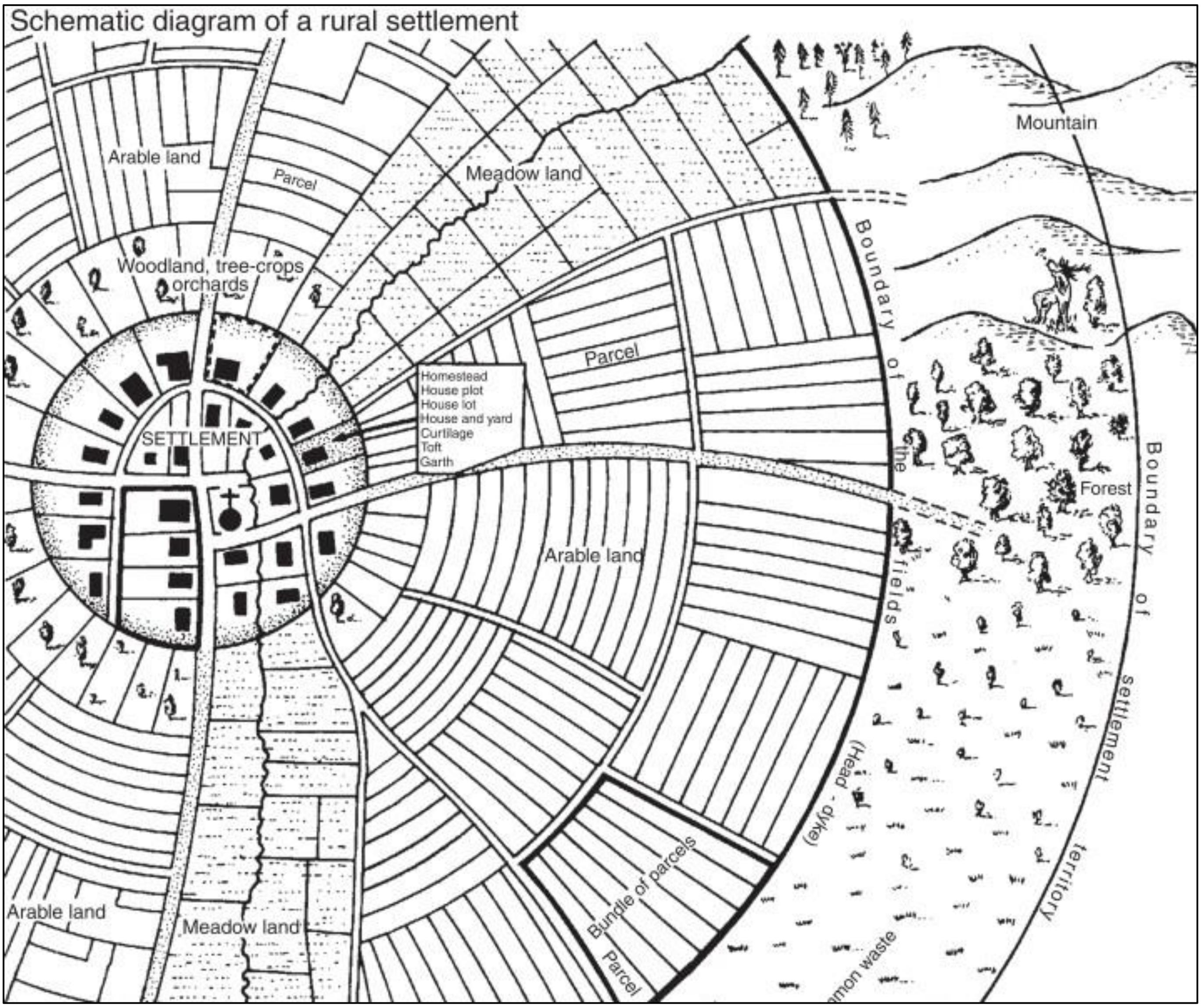
3.2.4. Large scale solar farms would best be built in an urban area.

3.2.5. One of the causes of rural-urban migration is an increase in yield per unit area of agricultural land.

(5x1)(5)

Question 3.3 **Rural Settlements**

Refer to the diagram below and answer the questions that follow.



3.3.1. Is this rural settlement likely to be engaging in commercial or subsistence farming? (1x1) (1)

3.3.2. Justify your answer to QUESTION 3.3.1. with TWO pieces of evidence from the diagram. (2x2) (4)

3.3.3. This settlement developed on a wet-point site. Define the term *wet-point site.* (1x2) (2)

3.3.4. List THREE site factors that would have likely contributed to the development of this settlement. (3x1) (3)

3.3.5. Define the concept of a settlement's *situation.* (1x2) (2)

3.3.6. Is this settlement's situation likely to be positive or negative? Justify your answer. (1+2) (3)

[15]

Question 3.4 **Rural Settlement Issues**

Refer to the article below and answer the questions that follow.

Rural healthcare in South Africa: A tale of adversity and heroism

Although the state of rural healthcare in South Africa is fraught with challenges, including a shortage of staff, political interference, and a lack of resources, it is a story of adversity and heroism. Despite these challenges, there are a few islands of rural excellence where healthcare professionals have shown innovation and resilience, including the invention of apps to streamline referrals and the transformation of rural hospitals.

Heroism in adversity – SA's heart-rending rural health story

Public sector rural healthcare in South Africa is politically mismanaged, understaffed, underfunded and under-supervised, with unions reigning supreme, intimidating competent and suitably qualified leaders into shunning the top hospital jobs.

In an exclusive interview with Medbrief Africa, National Health Ombudsman, Professor Makgoba put abysmal public health care delivery in all but the Western Cape, Limpopo and KwaZulu Natal ("where they have at least found a little direction") down to dismal provincial and hospital leadership, infrastructural decay, and an almost universal lack of human resource capacity.

"Because of understaffing at every level, you have little institutional knowledge about the disciplines being practised in those hospitals. Everything has suffered, and everyone is overworked and overstretched. Seniors become irritated with the human resource leadership and infrastructure of the hospital," he adds.

Under 10% of graduates go rural.
There are very few islands of rural excellence left in a sea of otherwise ideologically correct, semi-functional provincial health administrations. Despite a change in funding policy to get more money into rural areas (38 of the 52 health districts are considered rural), the money is almost always poorly or wastefully managed, while just 200 of the estimated annual 2 500 medical graduates are employed in State hospitals, (post community service), often not in rural areas – a function of ever-diminishing budgets.

Administrative 'guile'.

In KwaZulu Natal, a new healthcare staff establishment norm was set in 2013 – but has never been implemented or translated into the Persal (human resources and payroll) system. This means the true vacancy rates for healthcare professionals, cleaners, kitchen staff, porters and admin staff are impossible to establish. The current standing instructions from KZN Department of Health are that if you cannot fill a post, you abolish it – a slashing of staff numbers via administrative guile.

“So, the vacancy rate will show as, say, twenty percent when it’s actually sixty percent,” Hobe adds.

Luckily, South Africa has its medical heroes and tireless advocates and activists who see the half-full glass. As Ben Gaunt, whose book, “Hope, A Goat and a Hospital”, chronicles his core team’s remarkable Zithulele journey, signs off on his e-mails, “If you don’t have a dream, how will you ever have a dream come true?”

*Guile - sly or cunning intelligence.

3.4.1. What rural issue is being highlighted in the article? (1x1) (1)

3.4.2. List TWO reasons for this issue as mentioned in the text. (2x1) (2)

3.4.3. Explain why it is challenging to get doctors working in rural areas. (2x2) (4)

3.4.4. In a short paragraph of approximately 8 lines, discuss how you would solve the issue highlighted in the article. (4x2) (8)

[15]

QUESTION 3 TOTAL: 40 MARKS

TOTAL SECTION A: 120 Marks

SECTION B: MAPWORK

RESOURCE MATERIAL:

1. An extract from topographic map 2931CA VERULAM
2. Orthophoto map 2731 CA 11 VERULAM

INSTRUCTIONS & INFORMATION

1. Answer ALL the questions on your answer sheet.
2. You are provided with a 1:50 000 topographic map (**2931CA VERULAM**) and an orthophoto map (2931 CA 11 VERULAM) which is part of a mapped area.
3. You must hand in both the topographic map and the orthophoto map to the invigilator at the end of this examination session.
4. Do not mark the maps.
5. Show all calculations and formulae, where applicable. Marks will be allocated for these.
6. Indicate the unit of measurement in the working out and final answer of calculations,

   e.g. 10 km, 2.1 cm.
7. You may use a non-programmable calculator.
8. You may use a magnifying glass.
9. The area demarcated in RED on the topographic map represents the area covered by the orthophoto map.

GENERAL INFORMATION OF VERUMAM

Verulam at a glance

Verulam is a town situated towards the northern regions of the thriving metropolitan area of Durban. A gateway to the pristine beaches and farmlands of the North Coast a highly regarded holiday and agricultural region of KwaZulu-Natal. Known area for farming maize and sugar cane.

There is a farmers market that happens daily and even though Verulam is considered to be quite a sleepy place. Their daily market takes place until 4 pm and has people from all over the province frequenting the small town to buy fresh produce and other types of goods and products. The Verulam Market has had a huge role to play in the development of the area and has been in existence since the 1800's.

QUESTION 4: MAP SKILLS AND CALCULATIONS

The questions below are based on the 1:50 000 topographic map (2931CA VERULAM), as well as the orthophoto map as part of the mapped area. Various options are provided as possible answers to the following questions. Write down the question number and the letter (A-D) only.

4.1.1 The co-ordinates of the Zwolle Estate at **H** (A 2/ B2) are approximately…

A 31^0 36’ 0” S 29^0 03’ 37” E

B 29^0 36’ 0” S 31^0 03’ 37” E

C 31^0 36’ 37” S 29^0 03’ 0” E

D 29^0 36’ 37” S 31^0 03’ 0” E

4.1.2 The topographic map code directly West of 29 31 CA is…

A 29 31 DD

B 29 30 DB

C 29 31 DB

D 29 30 DD

4.1.3 On the orthophotograph, the gradient between spot height 29 (**A5**) and spot height 133 (**D2**) is ………..and is classified as a ……….. slope.

i 1: 76.9

ii 1: 15.8

iii steep

iv gentle

A i and iii

B ii and iii

C i and iv

D ii and iv

4.1.4 The scale of the topographic map is 5 times … than the orthophoto, and the topographic map covers a … area.

A larger; smaller

B larger; larger

C smaller; larger

D smaller; smaller (4x1) (4)

4.1.5 On the topographical map, work out the magnetic bearing of the benchmark 52.8 (**E5**)

from trig beacon 70 (**C3**).

Difference in years: 7 years

Total annual change: 9' x 7 = 63' (1^0 03'W)

Magnetic Bearing = True Bearing + Magnetic Declination (4x1) (4)

4.1.6 Calculate the area of the demarcated red orthophoto, on the topographical map, in **km$^{2.}$**
Use the following length distance: 3.7 cm = 1.85km

Formula: Area = Length x Breadth (2x1) (2)

[10]

4.2 APPLICATION & INTERPRETATION

The questions below are based on the 1:50 000 topographic map (2931CA VERULAM), as well as the orthophoto map as part of the mapped area. Various options are provided as possible answers to the following questions. Write down the question number and the letter (A-D) only.

4.2.1 The river feature found at **J** is the result of …., on the topographical map.

A erosion

B deposition

C undercutting

D abstraction

4.2.2 The King Shakas landing strips (**C5**) are running………………..due to the strong ……..winds

i North west to southeast

ii South west to northeast

iii South easterly

iv North westerly

A i and iii

B ii and iii

C i and iv

D ii and iv (2x1) (2)

4.2.3 Refer to Bellamont (**E5**) on the topographical map.

a) Is the settlement pattern at Bellamont *dispersed* or *nucleated*? (1x1) (1)

b) State TWO site factors that favoured the development of this settlement. (2x1) (2)

c) Provide the situation of Bellamont? (1x2) (2)

4.2.4 Bellamont settlement is linked to commercial maize farming.

a) Would this type of farming be *commercial* or *subsistence*? (1x1) (1)

b) Support your answer in QUESTION 3.2.4 a, using evidence from the map. (1x2) (2)

4.2.5 Zwolle Estate at **H** (**A/B2**) would be classified as *dry point* settlement.

Give a reason to support this statement. (1x2) (2)

[12]

4.3 GEOGRAPHICAL INFORMATION SYSTEMS

Refer to the information below and answer the questions that follow.

About this app

As quick warning measures and improved technologies are more at demand for improved maize production and food security, the ARC (Agricultural Research Council) is pleased to introduce the maize producers to new and improved mobile app that comprises of the latest collection of scientifically proven data on maize production, which can be easily downloaded for quick and updated information on the go.

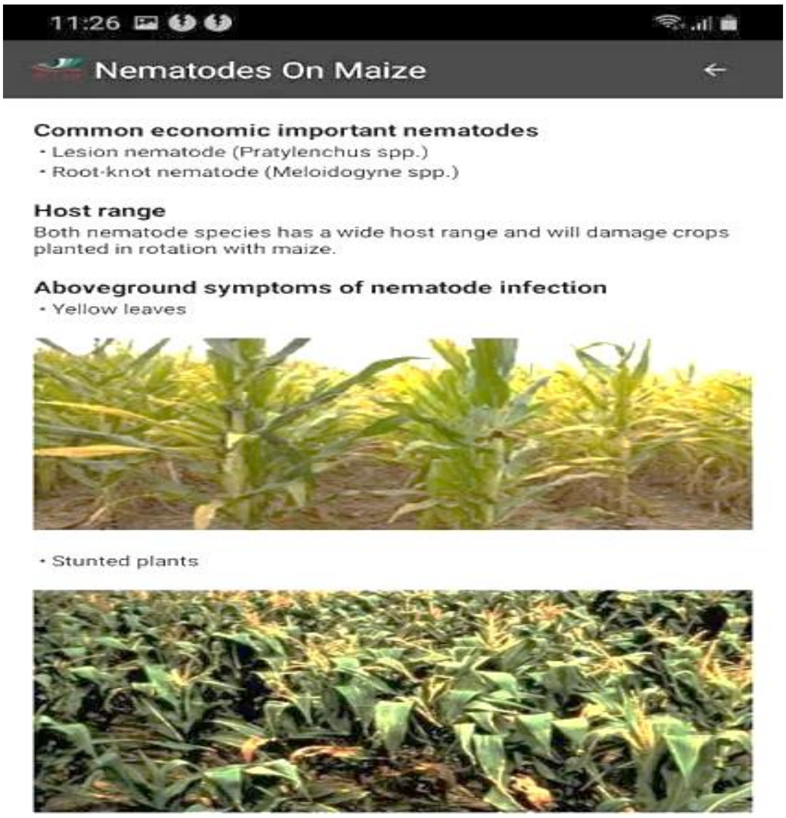
The ARC (Agricultural Research Council) remains committed in providing the best technologies to the maize farmers

GIS is made up of many different components, such as humans, hardware and software.

4.3.1 What type of component will ARC.LNR be part of? (1x1) (1)

4.3.2 List TWO different types of data that ARC.LNR could offer the farmers? (2x1) (2)

4.4 The following information was taken from the ARC.LNR farming site.



4.4.1 Classify the above photograph as either primary or secondary data. (1x1) (1)

4.4.2 Resolution is an important factor when taking photographs.

a) Define the term *resolution*. (1x2) (2)

b) How would a higher resolution in the photographs assist a GIS specialist to find a solution for the environmental issue depicted. (1x2) (2)

[8]

QUESTION 4 TOTAL: 30 MARKS

TOTAL SECTION B: 30 Marks

<u>GRAND TOTAL: 150 Marks</u>